

The excess-kurtosis power law

Tide retrospective, laplace seabed

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1 Setting

Two earlier tides on the laplace monomial track gave the excess kurtosis $\langle x^4 \rangle - 3\langle x^2 \rangle^2$ of the symmetric weight $\exp(-t x^{2k}/(2k)!)$ in closed form, and packaged the even moment $\langle x^{2j} \rangle_t$ as an asymptotic power law. This tide closes the small remaining gap: the asymptotic form of the excess kurtosis itself.

2 The thing formalised

For $k \geq 1$ and every $t > 0$,

$$\langle x^4 \rangle - 3\langle x^2 \rangle^2 = K(k) \cdot t^{-2/k}, \quad K(k) := (2k)!^{2/k} \left(\frac{\Gamma(5/(2k))}{\Gamma(1/(2k))} - 3 \left(\frac{\Gamma(3/(2k))}{\Gamma(1/(2k))} \right)^2 \right),$$

hence $\sim K(k) t^{-2/k}$ at atTop. The Lean additions are `kurtosisConst` and the three rungs `_eq.const_mul_rpow`, `_rescaled_tendsto`, `_isEquivalent_rpow`. At $k = 1$ the constant $K(1) = 0$: a Gaussian’s excess kurtosis, and its scaling, are trivially zero.

3 Proof strategy

The key observation is that the excess kurtosis, though a difference of two moments, involves two even moments whose t -powers coincide — both scale as $((2k)!/t)^{2/k}$ — so the difference is still a single pure power $K(k) t^{-2/k}$, not a genuine two-term asymptotic. The closed form from the earlier tide already exposes that common factor; `Real.div_rpow` peels $((2k)!/t)^{2/k}$ into $(2k)!^{2/k} t^{-2/k}$ and `ring` finishes. The rescaled `Tendsto` and the `IsEquivalent` rungs are the constant-sequence and reflexivity arguments used verbatim for the moment asymptotic.

4 Roadblocks and resolutions

The mathematics compiled on the first attempt. The only friction was a Mathlib `style.header` linter warning (“Copyright too short!”) that surfaces on fresh elaboration when the copyright block omits an `Authors:` line; adding one (the Mathlib-standard header form) silenced it. Most existing seabed files predate this linter and carry the warning latently.

5 What was Mathlib, what was new

Only the `Real.rpow` algebra and the `IsEquivalent/Tendsto` API came from Mathlib. The excess-kurtosis power law and its constant are new, completing the asymptotic form of the connected four-point function on the monomial track. No GPT consult was needed.